

YEAR 7 MATHEMATICS

Chapter 8

Area and volume

MIXED PRACTICE WORKSHEETS



8

How to use this booklet

- Each lesson is **one double-sided sheet**. Write your name, class and date at the top.
- Then work down the **columns**, left to right. They get harder as you go: **Easy** **Easy** **Medium** on page 1, then **Medium** **Challenging** **Challenging** on page 2.
- Most questions practise **today's skill**. A question marked **from 1.03** comes from an earlier lesson, to keep it fresh. Stuck? Look back at that lesson.
- Each question tells you what to do: read it carefully. Show your working on the lines. The first question with working in each Easy column shows you how to set it out.

WE DO

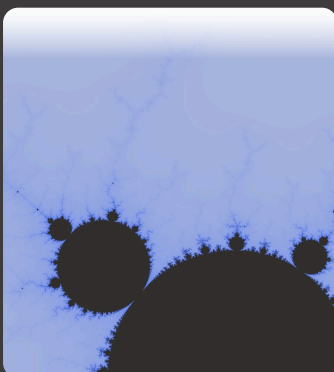
The grey box at the top. Your teacher shows the key idea with two or three examples. **Copy every step** into the space.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
8.01	Units of measurement	3		☐
8.02	Perimeter	5		☐
8.03	Circumference of a circle	7		☐
8.04	Area units	9		☐
8.05	Area of squares and rectangles	11		☐
8.06	Area of a triangle	13		☐
8.07	Area of a parallelogram	15		☐
8.08	Area of composite shapes	17		☐
8.09	Prisms, cross-sections and views	19		☐
8.10	Volume units	21		☐
8.11	Volume of a rectangular prism	23		☐
8.12	Volume of a prism	25		☐
8.13	Capacity	27		☐

In this chapter you will

- convert units of length, mass, capacity and time
- find perimeter and circumference
- convert area units and find the area of rectangles, triangles, parallelograms and composite shapes
- draw cross-sections and views of prisms
- convert volume units and find the volume of prisms
- find capacity in litres



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the thin branches (dendrites) that grow from the very top of the set. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

NAME _____ CLASS _____ DATE _____

WE DO
KEY IDEA

To change to a smaller unit, multiply: there are more of them. To change to a larger unit, divide. 1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm, 1 kg = 1000 g, 1 L = 1000 mL.

Watch your teacher.
Copy every step.

E1 Change 3.5 km to metres.

E2 Change 450 g to kilograms.

E3 Add 2 m and 35 cm. Give the answer in centimetres.

Easy

Questions 1–9

1 Change 23 000 mm to m.

2 Change 51 L to mL.

3 Change 4 hours to minutes.

4 Change 21 km to m.

5 Change 38 kg to g.

6 Change 2 minutes to seconds.

7 Change 3 200 cm to m.

8 Change 3 000 g to kg.

9 $58 \div 10$

from 7.04

Easy

Questions 10–18

10 Change 30 000 L to kL.

11 Change 5 minutes to seconds.

12 Change 51 cm to mm.

13 What is the order of rotational symmetry of the capital letter I?

from 6.04

14 Change 3.1 m to mm.

m to mm: $\times$ ____

 $3.1 \times 1\,000 =$ _____

15 Change 16 000 L to kL.

16 Change 8 minutes to seconds.

17 $46 \div 10$

from 7.04

18 Change 32 km to m.

Medium

Questions 19–26

19 Change 24 000 L to kL.

20 Add 5 m and 605 cm. Give the answer in metres.

21 Change 3.45 km to m.

from 7.04

22 Change 4 h 17 min to minutes.

23 Change 36 km to m.

24 Solve $3p + 7 = 10$.

from 5.12

25 Add 7 m and 737 cm. Give the answer in metres.

26 Change 2 h 49 min to minutes.

Medium

Questions 27–37

- 27 Add 4 m and 584 cm. Give the answer in metres.

- 28 Change 1 h 54 min to minutes.

- 29 $83.6 \div 1\,000$

from 7.04

- 30 Change 2.7 kL to L.

- 31 Find the missing number:
 $9.07 \times \square = 907$

from 7.04

- 32 A 6 L jug has 311 mL poured out. How much is left, in mL?

- 33 Add 2 m and 328 cm. Give the answer in metres.

- 34 Order from largest to smallest:
0.09, 0.2, 0.699, 0.9

from 7.01

- 35 Change 2 h 13 min to minutes.

- 36 $6.47 \times 1\,000$

from 7.04

- 37 Change 33 m to cm.

Challenging

Questions 38–47

- 38 A 7 L jug has 258 mL poured out. How much is left, in mL?

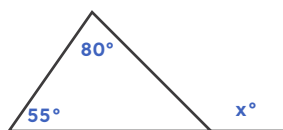
- 39 Order from shortest to longest: 121 cm, 1 060 mm, 1.12 m

- 40 Change 77 m to km.

from 7.04

- 41 8 packets of rice weigh 245 g each. Find the total mass in kilograms.

- 42 Find x , the exterior angle.



from 6.07

- 43 A 6 L jug has 243 mL poured out. How much is left, in mL?

- 44 Order from shortest to longest: 1 026 mm, 1.05 m, 108 cm

- 45 $10 \div 100$

from 7.04

- 46 7 packets of rice weigh 365 g each. Find the total mass in kilograms.

- 47 Change 0.28 L to mL.

from 7.04

Challenging

Questions 48–57

- 48 Order from shortest to longest: 1.16 m, 1 128 mm, 121 cm

- 49 11 packets of rice weigh 243 g each. Find the total mass in kilograms.

- 50 Solve $\frac{t}{9} = 2$.

from 5.11

- 51 A 5 L jug has 163 mL poured out. How much is left, in mL?

- 52 $8.3 \div 1\,000$

from 7.04

- 53 Change 4 h 42 min to minutes.

- 54 Order from shortest to longest: 1.17 m, 1 147 mm, 124 cm

- 55 Find the missing number:
 $5.89 \times \square = 589$

from 7.04

- 56 5 packets of rice weigh 171 g each. Find the total mass in kilograms.

- 57 Write $\frac{96}{100}$ as a decimal.

from 7.02

NAME _____ CLASS _____ DATE _____

WE DO

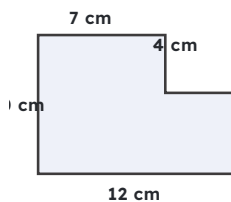
Watch your teacher.
Copy every step.

KEY IDEA

Perimeter is the distance around the outside of a shape: add all the side lengths. For a rectangle, $P = 2l + 2w$. Find any missing sides first.

- E1** Find the perimeter of an 8 m by 5 m rectangle.

- E2** Find the perimeter. (Not to scale.)



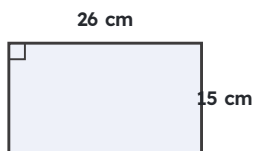
- E3** A square has a perimeter of 36 cm. Find a side.

Easy

Questions 1–7

- 1** Find the perimeter of an 8 m by 4 m rectangle.

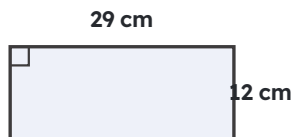
- 2** Find the perimeter.



- 3** Find the perimeter of a square with sides of 19 cm.

- 4** Find the perimeter of a 17 m by 4 m rectangle.

- 5** Find the perimeter.



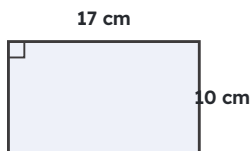
- 6** Find the perimeter of a square with sides of 8 cm.

- 7** Find the perimeter of a 15 m by 2 m rectangle.

Easy

Questions 8–12

- 8** Find the perimeter.



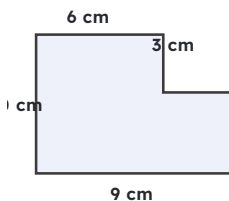
- 9** Find the perimeter of a square with sides of 6 cm.

- 10** Find the perimeter of a 10 m by 3 m rectangle.

- 11** Round 11.5 to the nearest whole number.

from 7.10

- 12** Find the perimeter. Find the missing sides first. (Not to scale.)

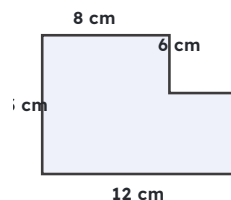


missing: ____ and ____

Medium

Questions 13–17

- 13** Find the perimeter. Find the missing sides first. (Not to scale.)



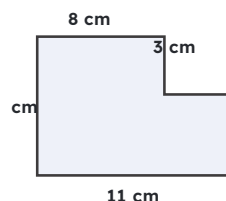
- 14** Find the perimeter of a regular octagon with sides of 14 cm.

- 15** Change 56 cm to mm.

from 8.01

- 16** A triangle has sides 7 cm, 5 cm and 6 cm. Find its perimeter.

- 17** Find the perimeter. Find the missing sides first. (Not to scale.)



Medium

Questions 18–24

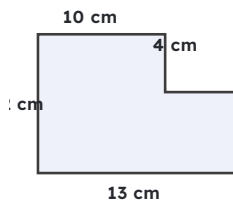
- 18 Find the perimeter of a regular octagon with sides of 4 cm.

- 19 A triangle has sides 7 cm, 4 cm and 9 cm. Find its perimeter.

- 20 Change 23 000 kg to t.

from 8.01

- 21 Find the perimeter. Find the missing sides first. (Not to scale.)



- 22 Change 1 h 46 min to minutes.

from 8.01

- 23 A rectangle has a perimeter of 66 m and a length of 27 m. Find its width.

- 24 Find the perimeter of a regular octagon with sides of 8 cm.

Challenging

Questions 25–32

- 25 A rectangle has a perimeter of 58 m and a length of 16 m. Find its width.

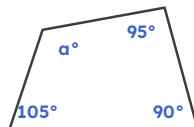
- 26 An 11 m by 6 m paddock is fenced. Fencing costs \$16 a metre. Find the cost.

- 27 Add 4 m and 518 cm. Give the answer in metres.

from 8.01

- 28 A square and an equilateral triangle each have a perimeter of 144 cm. Find each side length.

- 29 Find a . (Not to scale.)



- 30 A rectangle has a perimeter of 82 m and a length of 30 m. Find its width.

from 6.09

- 31 A 28 m by 8 m paddock is fenced. Fencing costs \$22 a metre. Find the cost.

- 32 Change 3 hours to minutes.

from 8.01

Challenging

Questions 33–41

- 33 A 24 m by 17 m paddock is fenced. Fencing costs \$16 a metre. Find the cost.

- 34 A square and an equilateral triangle each have a perimeter of 120 cm. Find each side length.

- 35 How many axes of symmetry does the word MUM have (in capitals)?

from 6.03

- 36 A rectangle has a perimeter of 76 m and a length of 30 m. Find its width.

- 37 Change 26 000 g to kg.

from 8.01

- 38 A triangle has sides 3 cm, 9 cm and 8 cm. Find its perimeter.

- 39 A 16 m by 10 m paddock is fenced. Fencing costs \$10 a metre. Find the cost.

- 40 Change 4 h 7 min to minutes.

from 8.01

- 41 A square and an equilateral triangle each have a perimeter of 96 cm. Find each side length.

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

The circumference is the distance around a circle: $C = \pi d$, or $C = 2\pi r$. Use the π key on your calculator, then round as asked.

Watch your teacher.
Copy every step.

E1 Find the circumference when $d = 10$ cm (1 decimal place).

E2 Find the circumference when $r = 6$ m (1 decimal place).

E3 A circle has a circumference of 50 cm. Find its diameter.

Easy

Questions 1–8

1 Find the circumference of a circle with diameter 20 cm (1 decimal place).

2 A circle has a radius of 7 m. What is its diameter?

3 Find the circumference of a circle with radius 12 cm (1 decimal place).

4 Find the circumference of a circle with diameter 34 cm (1 decimal place).

5 A circle has a radius of 28 m. What is its diameter?

6 Find the circumference of a circle with radius 6 cm (1 decimal place).

7 Find the circumference of a circle with diameter 9 cm (1 decimal place).

8 A circle has a radius of 9 m. What is its diameter?

Easy

Questions 9–16

9 A circle has a radius of 10 m. What is its diameter?

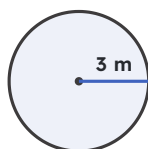
10 Find the circumference of a circle with radius 5 cm (1 decimal place).

11 Find the circumference of a circle with diameter 28 cm (1 decimal place).

12 Given $30 \times 7 = 210$, find 30×0.7 .

from 7.05

13 Find the circumference (1 decimal place).



$$C = ___ \times \pi \times ___$$

$$\approx ___ \text{ m}$$

14 A circle has a radius of 24 m. What is its diameter?

15 Find the circumference of a circle with radius 13 cm (1 decimal place).

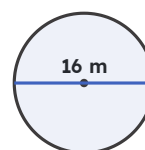
16 Round 46.2 to the nearest whole number.

from 7.10

Medium

Questions 17–22

17 Find the circumference (1 decimal place).

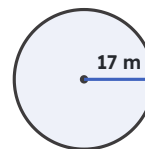


18 A bike wheel has a diameter of 44 cm. How far does it roll in one turn (1 decimal place)?

19 Add 6 m and 862 cm. Give the answer in metres.

from 8.01

20 Find the circumference (1 decimal place).



21 A bike wheel has a diameter of 58 cm. How far does it roll in one turn (1 decimal place)?

22 $48.7 \div 1\,000$

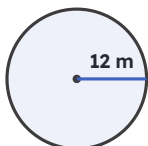
from 7.04

Medium

Questions 23–30

- 23 A bike wheel has a diameter of 42 cm. How far does it roll in one turn (1 decimal place)?

- 24 Find the circumference (1 decimal place).



- 25 Find the perimeter of a square with sides of 11 cm.

from 8.02

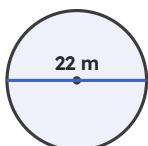
- 26 A circle has a circumference of 88 cm. Find its diameter (1 decimal place).

- 27 Write $\frac{1}{6}$ as a decimal rounded to 2 decimal places.

from 7.10

- 28 A bike wheel has a diameter of 80 cm. How far does it roll in one turn (1 decimal place)?

- 29 Find the circumference (1 decimal place).



- 30 True or false? Every square is a rectangle.

from 6.08

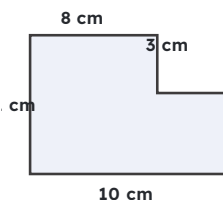
Challenging

Questions 31–38

- 31 A circle has a circumference of 82 cm. Find its diameter (1 decimal place).

- 32 Find the perimeter of a semicircle with diameter 19 cm (1 decimal place).

- 33 Find the perimeter. Find the missing sides first. (Not to scale.)



from 8.02

- 34 A wheel of diameter 53 cm turns 49 times. How far does it travel, in metres (1 decimal place)?

- 35 $4 \div 0.5$

from 7.08

- 36 A circle has a circumference of 35 cm. Find its diameter (1 decimal place).

- 37 Find the perimeter of a semicircle with diameter 26 cm (1 decimal place).

- 38 Round 27.34 to 1 decimal place.

from 7.10

Challenging

Questions 39–46

- 39 Find the perimeter of a semicircle with diameter 12 cm (1 decimal place).

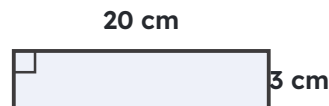
- 40 A wheel of diameter 67 cm turns 49 times. How far does it travel, in metres (1 decimal place)?

- 41 Solve $\frac{m}{4} + 8 = 17$.

from 5.12

- 42 A circle has a circumference of 25 cm. Find its diameter (1 decimal place).

- 43 Find the perimeter.



from 8.02

- 44 A bike wheel has a diameter of 52 cm. How far does it roll in one turn (1 decimal place)?

- 45 Find the perimeter of a semicircle with diameter 22 cm (1 decimal place).

- 46 Round \$8.486 to the nearest cent.

from 7.10

NAME _____ CLASS _____ DATE _____

WE DO**KEY IDEA**

$1 \text{ cm}^2 = 100 \text{ mm}^2$ (10×10). $1 \text{ m}^2 = 10\,000 \text{ cm}^2$ (100×100). $1 \text{ ha} = 10\,000 \text{ m}^2$. To change to a smaller unit, multiply; to a larger unit, divide.

Watch your teacher.
Copy every step.

E1 Change 3 cm^2 to mm^2 .

E2 Change $25\,000 \text{ cm}^2$ to m^2 .

E3 Change 4.5 ha to m^2 .

Easy

Questions 1–9

1 Change 29 cm^2 to mm^2 .

2 Change $50\,000 \text{ m}^2$ to ha .

3 How many mm^2 are in 1 cm^2 ?

4 Change 24 cm^2 to mm^2 .

5 Change $1\,900 \text{ mm}^2$ to cm^2 .

6 How many m^2 are in 1 hectare?

7 Change 20 m^2 to cm^2 .

8 Change $3\,000 \text{ mm}^2$ to cm^2 .

9 Change $39\,000 \text{ kg}$ to t .

from 8.01

Easy

Questions 10–18

10 Change $90\,000 \text{ m}^2$ to ha .

11 How many cm^2 are in 1 m^2 ?

12 Change 30 ha to m^2 .

13 A shape is turned 180° clockwise, then 180° clockwise. What single turn does the same?

from 6.02

14 Change 3.5 cm^2 to mm^2 . $1 \text{ cm}^2 = \underline{\hspace{2cm}}$ $3.5 \times 100 = \underline{\hspace{2cm}}$ **15** Change $20\,000 \text{ cm}^2$ to m^2 .

16 Change 14 ha to m^2 .

17 A circle has a radius of 5 m . What is its diameter?

from 8.03

18 Change 5 m^2 to cm^2 .

Medium

Questions 19–26

19 Change 8.5 m^2 to cm^2 .

20 Change $8\,900 \text{ mm}^2$ to cm^2 .

21 Find the perimeter of a regular octagon with sides of 15 cm .

from 8.02

22 Change 7.5 m^2 to cm^2 .

23 Change $9\,100 \text{ mm}^2$ to cm^2 .

24 A rhombus has a side of 4 cm . Find its perimeter.

from 6.10

25 Change 8 km^2 to m^2 .

26 Change $8\,700 \text{ mm}^2$ to cm^2 .

Medium

Questions 27–36

- 27 Change $3\,500\text{ mm}^2$ to cm^2 .

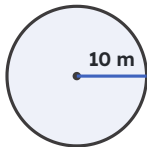
- 28 Change 9.5 m^2 to cm^2 .

- 29 Change 5 hours to minutes.

from 8.01

- 30 A paddock is 800 m by 200 m. Find its area in hectares.

- 31 Find the circumference (1 decimal place).



from 8.03

- 32 Change $6\,700\text{ mm}^2$ to cm^2 .

- 33 Change 4 cm^2 to mm^2 .

- 34 True or false? Every square is a rectangle.

from 6.08

- 35 A paddock is 500 m by 500 m. Find its area in hectares.

- 36 Find the perimeter of a square with sides of 6 cm.

from 8.02

Challenging

Questions 37–45

- 37 A paddock is 200 m by 400 m. Find its area in hectares.

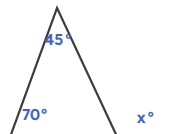
- 38 Which is larger: 0.8 m^2 or $6\,000\text{ cm}^2$?

- 39 Change 41 cm to mm.

from 8.01

- 40 A rug is 47 cm by 33 cm. Find its area in m^2 .

- 41 Find x , the exterior angle.



from 6.07

- 42 A paddock is 300 m by 200 m. Find its area in hectares.

- 43 Which is larger: 0.6 m^2 or $5\,000\text{ cm}^2$?

- 44 Find the circumference of a circle with diameter 7 cm (1 decimal place).

from 8.03

- 45 A rug is 51 cm by 26 cm. Find its area in m^2 .

Challenging

Questions 46–54

- 46 Which is larger: 0.6 m^2 or $7\,000\text{ cm}^2$?

- 47 A rug is 77 cm by 39 cm. Find its area in m^2 .

- 48 An isosceles triangle has an apex angle of 28° . Find each base angle.

from 6.06

- 49 A paddock is 600 m by 200 m. Find its area in hectares.

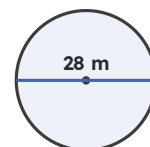
- 50 Change 5 L to mL.

from 8.01

- 51 Change $6\,300\text{ mm}^2$ to cm^2 .

- 52 Which is larger: 0.5 m^2 or $7\,000\text{ cm}^2$?

- 53 Find the circumference (1 decimal place).



from 8.03

- 54 A rug is 42 cm by 26 cm. Find its area in m^2 .

NAME _____ CLASS _____ DATE _____

WE DO

KEY IDEA

Area of a rectangle = length $\times$ width. Area of a square = side $\times$ side. Area is measured in square units, such as cm^2 and m^2 .

Watch your teacher.
Copy every step.

E1 Find the area of an 8 m by 5 m rectangle.

E2 Find the area of a square with sides of 7 cm.

E3 A rectangle has an area of 60 cm^2 and a width of 5 cm. Find its length.

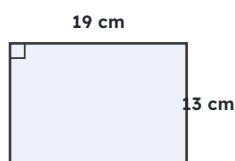
Easy

Questions 1–8

1 Find the area of a 9 m by 3 m rectangle.

2 Find the area of a square with sides of 8 cm.

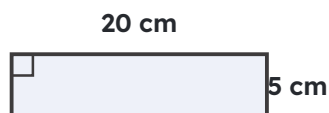
3 Find the area.



4 Find the area of an 8 m by 2 m rectangle.

5 Find the area of a square with sides of 6 cm.

6 Find the area.



7 Find the area of a 14 m by 13 m rectangle.

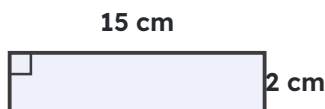
8 Find the area of a square with sides of 11 cm.

Easy

Questions 9–15

9 Find the area of a square with sides of 12 cm.

10 Find the area.



11 Find the area of a 12 m by 12 m rectangle.

12 $8.8 - 1.9$

from 7.03

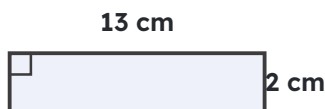
13 A rectangle has an area of 72 m^2 and a width of 12 m. Find its length.

$$\underline{\quad} = l \times \underline{\quad}$$

$$l = \underline{\quad} \div \underline{\quad} = \underline{\quad} \text{ m}$$

14 Find the area of a square with sides of 5 cm.

15 Find the area.



Medium

Questions 16–22

16 A rectangle has an area of 252 m^2 and a width of 14 m. Find its length.

17 Find the area of a 4 m by 7 m rectangle.

18 Add 2 m and 466 cm. Give the answer in metres.

from 8.01

19 A square has an area of 25 cm^2 . How long is each side?

20 A rectangle has an area of 120 m^2 and a width of 10 m. Find its length.

21 A bike wheel has a diameter of 72 cm. How far does it roll in one turn (1 decimal place)?

from 8.03

22 Find the area of a 3.4 m by 6 m rectangle.

Medium

Questions 23–32

- 23 Find the area of an 8 m by 5 m rectangle.

- 24 A square has an area of 81 cm^2 . How long is each side?

- 25 A rhombus has a side of 14 cm. Find its perimeter.

from 6.10

- 26 A rectangle has an area of 154 m^2 and a width of 11 m. Find its length.

- 27 Change 9 minutes to seconds.

from 8.01

- 28 Carpet costs \$56 per m^2 . Find the cost to carpet an 8 m by 4 m room.

- 29 Find the area of a 3.2 m by 6 m rectangle.

- 30 Find the perimeter of a square with sides of 20 cm.

from 8.02

- 31 A square has an area of 121 cm^2 . How long is each side?

- 32 Change 3 cm^2 to mm^2 .

from 8.04

Challenging

Questions 33–40

- 33 Carpet costs \$37 per m^2 . Find the cost to carpet an 8 m by 6 m room.

- 34 How many 20 cm $\times$ 20 cm tiles cover a floor 6 m by 3 m?

- 35 Order from largest to smallest: 0.06, 0.18, 0.9, 0.447

from 7.01

- 36 An 8 m by 8 m lawn has a 1 m wide path around the outside. Find the area of the path.

- 37 Find the circumference of a circle with radius 20 cm (1 decimal place).

from 8.03

- 38 Carpet costs \$39 per m^2 . Find the cost to carpet an 8 m by 4 m room.

- 39 How many 20 cm $\times$ 20 cm tiles cover a floor 8 m by 3 m?

- 40 Change 420 mm to cm.

from 8.01

Challenging

Questions 41–49

- 41 How many 20 cm $\times$ 20 cm tiles cover a floor 5 m by 4 m?

- 42 A 9 m by 4 m lawn has a 1 m wide path around the outside. Find the area of the path.

- 43 Given $34 \times 7 = 238$, find 34×0.7 .

from 7.05

- 44 Carpet costs \$51 per m^2 . Find the cost to carpet a 5 m by 6 m room.

- 45 Change 2 500 mL to L.

from 8.01

- 46 A square has an area of 196 cm^2 . How long is each side?

- 47 How many 20 cm $\times$ 20 cm tiles cover a floor 6 m by 6 m?

- 48 Change 21 ha to m^2 .

from 8.04

- 49 A 12 m by 6 m lawn has a 1 m wide path around the outside. Find the area of the path.

NAME _____ CLASS _____ DATE _____

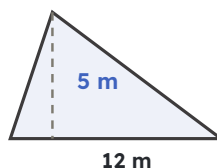
WE DO
KEY IDEA

Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height}$. The height is measured at right angles (perpendicular) to the base.

Watch your teacher.
Copy every step.

- E1** Find the area: base 10 cm, height 6 cm.

- E2** Find the area. (Not to scale.)



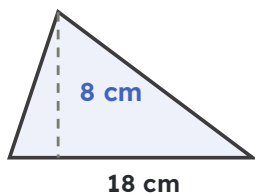
- E3** A triangle has an area of 24 cm^2 and a base of 8 cm. Find its height.

Easy

Questions 1–6

- 1** Find the area of a triangle with base 13 cm and height 14 cm.

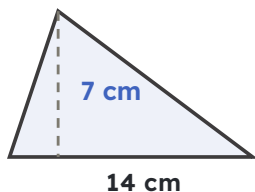
- 2** Find the area. (Not to scale.)



- 3** A rectangle is 4 cm by 8 cm. A diagonal cuts it into two triangles. Find the area of one triangle.

- 4** Find the area of a triangle with base 15 cm and height 10 cm.

- 5** Find the area. (Not to scale.)

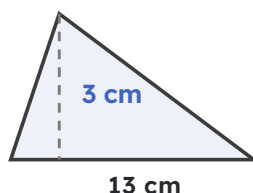


- 6** A rectangle is 11 cm by 6 cm. A diagonal cuts it into two triangles. Find the area of one triangle.

Easy

Questions 7–11

- 7** Find the area. (Not to scale.)



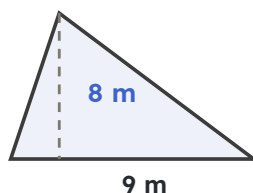
- 8** A rectangle is 3 cm by 9 cm. A diagonal cuts it into two triangles. Find the area of one triangle.

- 9** Find the area of a triangle with base 8 cm and height 6 cm.

- 10** $1.2 \div 0.6$

from 7.08

- 11** Find the area. Write the formula first. (Not to scale.)



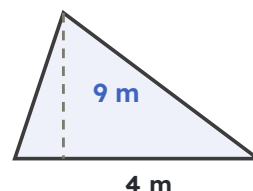
$$A = \frac{1}{2} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$$

$$= \underline{\hspace{1cm}} \text{ m}^2$$

Medium

Questions 12–15

- 12** Find the area. Write the formula first. (Not to scale.)



- 13** Find the area of a triangle with base 4 m and height 4 m.

- 14** Add 2 m and 154 cm. Give the answer in metres.

from 8.01

- 15** A right-angled triangle has shorter sides of 6 cm and 12 cm. Find its area.

Medium

Questions 16–22

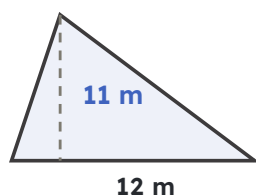
- 16 Find the area of a triangle with base 9.8 m and height 16 m.

- 17 A right-angled triangle has shorter sides of 4 cm and 8 cm. Find its area.

- 18 Find the change from \$20 for \$8.30.

from 7.11

- 19 Find the area. Write the formula first. (Not to scale.)



- 20 Find the perimeter.



from 8.02

- 21 A triangle has an area of 77 cm^2 and a base of 11 cm. Find its height.

- 22 Find the area of a triangle with base 5.5 m and height 6 m.

Challenging

Questions 23–31

- 23 A triangle has an area of 36 cm^2 and a base of 6 cm. Find its height.

- 24 Two triangles have the same base and height: 6 cm and 10 cm. One is right-angled, one is not. Compare their areas.

- 25 Write $\frac{1}{20}$ as a decimal.

from 7.02

- 26 A square of side 8 cm is cut along a diagonal. Find the area of each triangle.

- 27 Find the circumference of a circle with radius 14 cm (1 decimal place).

from 8.03

- 28 A triangle has an area of 52 cm^2 and a base of 8 cm. Find its height.

- 29 A square of side 9 cm is cut along a diagonal. Find the area of each triangle.

- 30 Change 4 h 36 min to minutes.

from 8.01

- 31 A triangle has an area of 152 cm^2 and a base of 19 cm. Find its height.

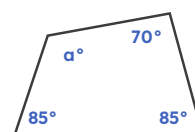
Challenging

Questions 32–40

- 32 A square of side 12 cm is cut along a diagonal. Find the area of each triangle.

- 33 A triangle has an area of 66 cm^2 and a base of 12 cm. Find its height.

- 34 Find a . (Not to scale.)



from 6.09

- 35 A right-angled triangle has shorter sides of 8 cm and 6 cm. Find its area.

- 36 Find the perimeter of a square with sides of 8 cm.

from 8.02

- 37 A square of side 4 cm is cut along a diagonal. Find the area of each triangle.

- 38 A triangle has an area of 90 cm^2 and a base of 20 cm. Find its height.

- 39 Change 26 ha to m^2 .

from 8.04

- 40 A right-angled triangle has shorter sides of 9 cm and 9 cm. Find its area.

NAME _____

CLASS _____

DATE _____

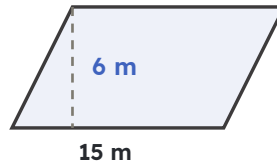
WE DO**KEY IDEA**

Area of a parallelogram = base $\times$ height. The height is perpendicular to the base: do not use the slanted side.

Watch your teacher.
Copy every step.

- E1** Find the area: base 9 cm, height 4 cm.
- _____
- _____
- _____
- _____

- E2** Find the area. (Not to scale.)



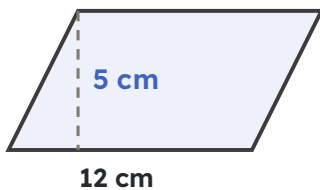
- E3** A parallelogram has an area of 56 m^2 and a height of 7 m. Find its base.
- _____
- _____
- _____
- _____

Easy

Questions 1–5

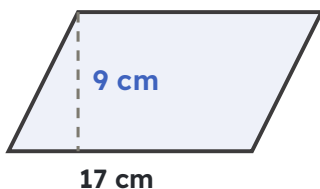
- 1** Find the area of a parallelogram with base 11 m and height 5 m.
- _____

- 2** Find the area. (Not to scale.)



- 3** Find the area of a parallelogram with base 12 m and height 13 m.
- _____

- 4** Find the area. (Not to scale.)

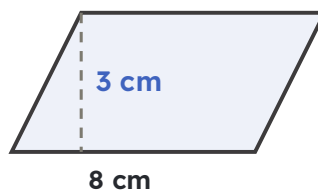


- 5** Find the area of a parallelogram with base 12 m and height 10 m.
- _____

Easy

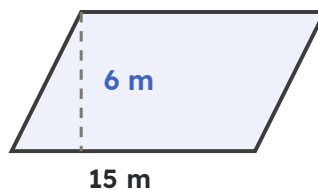
Questions 6–10

- 6** Find the area. (Not to scale.)



- 7** Find the area of a parallelogram with base 13 m and height 12 m.
- _____

- 8** Find the area. Write the formula first. (Not to scale.)



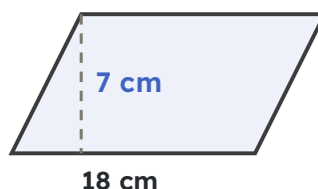
$$A = ___ \times ___$$

$$= ___ \text{ m}^2$$

- 9** Write $\frac{58}{100}$ as a decimal.

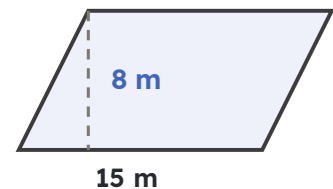
from 7.02

- 10** Find the area. (Not to scale.)

**Medium**

Questions 11–14

- 11** Find the area. Write the formula first. (Not to scale.)



- 12** A parallelogram has an area of 80 m^2 and a height of 8 m. Find its base.
- _____
- _____

- 13** Add 4 m and 418 cm. Give the answer in metres.

from 8.01

- 14** Find the area of a parallelogram with base 9.7 cm and height 6 cm.
- _____
- _____

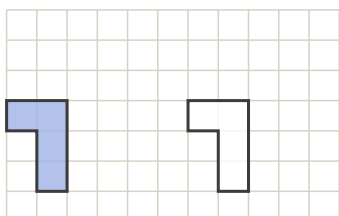
Medium

Questions 15–20

- 15 A parallelogram has an area of 21 m^2 and a height of 3 m. Find its base.

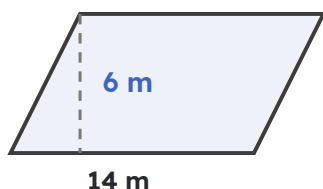
- 16 Find the area of a parallelogram with base 7.2 cm and height 8 cm.

- 17 Describe the translation from the shaded shape to the white shape.



from 6.01

- 18 Find the area. Write the formula first. (Not to scale.)



- 19 Find the perimeter.



from 8.02

- 20 A parallelogram has a base of 6 cm, a slanted side of 12 cm and a height of 10 cm. Find its area and its perimeter.

Challenging

Questions 21–28

- 21 A parallelogram has a base of 6 cm, a slanted side of 6 cm and a height of 4 cm. Find its area and its perimeter.

- 22 Why do we use the height, not the slanted side, for the area of a parallelogram?

- 23 Zac and friends share a \$15.90 bill equally between 2 people. How much each?

from 7.07

- 24 A parallelogram of base 12 m and height 4 m is cut along a diagonal. Find the area of each triangle.

- 25 How many m^2 are in 1 hectare?

from 8.04

- 26 A parallelogram has a base of 11 cm, a slanted side of 4 cm and a height of 3 cm. Find its area and its perimeter.

- 27 A rectangle and a parallelogram have the same base and the same height. Compare their areas.

- 28 A bike wheel has a diameter of 44 cm. How far does it roll in one turn (1 decimal place)?

from 8.03

Challenging

Questions 29–37

- 29 A parallelogram of base 5 m and height 7 m is cut along a diagonal. Find the area of each triangle.

- 30 A parallelogram has a base of 12 cm, a slanted side of 13 cm and a height of 10 cm. Find its area and its perimeter.

- 31 Classify a triangle with sides 11 cm, 11 cm, 11 cm by its sides.

from 6.05

- 32 Find the area of a parallelogram with base 3.4 cm and height 9 cm.

- 33 Change 9 hours to minutes.

from 8.01

- 34 A parallelogram of base 9 m and height 3 m is cut along a diagonal. Find the area of each triangle.

- 35 A parallelogram has a base of 7 cm, a slanted side of 8 cm and a height of 7 cm. Find its area and its perimeter.

- 36 Find the area of a 15 m by 12 m rectangle.

from 8.05

- 37 Find the area of a parallelogram with base 6 cm and height 2 cm.

NAME _____ CLASS _____ DATE _____

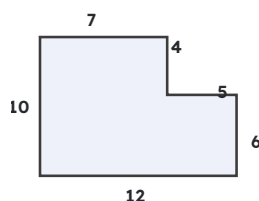
WE DO

KEY IDEA

Split a composite shape into rectangles and triangles, find each area, then add. Or find the area of a larger shape and subtract the part that is cut out.

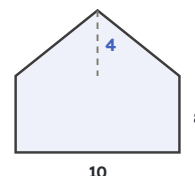
Watch your teacher.
Copy every step.

- E1** Find the area of the L-shape. (Not to scale.)



- E2** A 12 cm × 8 cm card has a 3 cm × 2 cm hole cut out. Find the area left.

- E3** Find the area of the shape. (Not to scale.)



Easy

Questions 1–6

- 1** A shape is made of a 6 m × 9 m rectangle and a 2 m × 8 m rectangle. Find its area.

$$\underline{\quad} + \underline{\quad}$$

$$= \underline{\quad} \text{ m}^2$$

- 2** A 9 cm × 9 cm card has a 3 cm × 3 cm hole cut out. Find the area left.

- 3** A shape is made of a 12 m × 2 m rectangle and a 4 m × 6 m rectangle. Find its area.

- 4** A 7 cm × 9 cm card has a 2 cm × 3 cm hole cut out. Find the area left.

- 5** A shape is made of an 11 m × 4 m rectangle and a 7 m × 6 m rectangle. Find its area.

- 6** A 9 cm × 6 cm card has a 1 cm × 1 cm hole cut out. Find the area left.

Easy

Questions 7–11

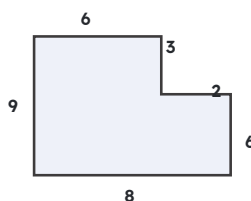
- 7** An 11 cm × 4 cm card has a 3 cm × 1 cm hole cut out. Find the area left.

$$\underline{\quad} - \underline{\quad}$$

$$= \underline{\quad} \text{ cm}^2$$

- 8** A shape is made of an 8 m × 3 m rectangle and a 9 m × 7 m rectangle. Find its area.

- 9** Find the area. Split it into two rectangles. (Not to scale.)



- 10** A number plus 8 is 24. What is the number?

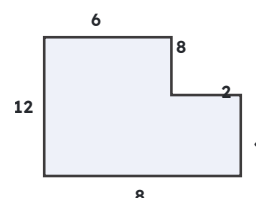
from 5.13

- 11** A 12 cm × 8 cm card has a 3 cm × 2 cm hole cut out. Find the area left.

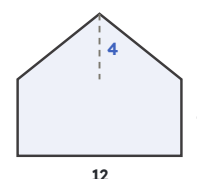
Medium

Questions 12–14

- 12** Find the area. Split it into two rectangles. (Not to scale.)



- 13** Find the area of the shape. (Not to scale.)



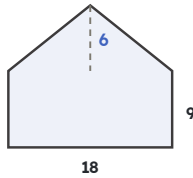
- 14** Add 2 m and 768 cm. Give the answer in metres.

from 8.01

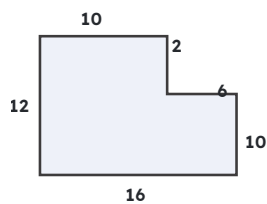
Medium

Questions 15–19

- 15 Find the area of the shape. (Not to scale.)



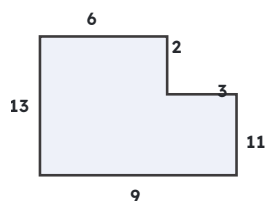
- 16 Find the area. Split it into two rectangles. (Not to scale.)



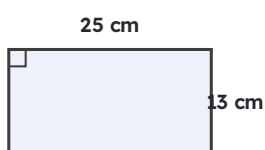
- 17 What is the order of rotational symmetry of the capital letter F?

from 6.04

- 18 Find the area two ways: by adding and by subtracting. (Not to scale.)



- 19 Find the perimeter.

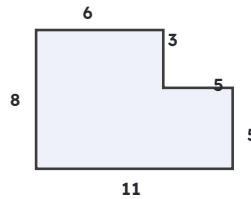


from 8.02

Challenging

Questions 20–25

- 20 Find the area two ways: by adding and by subtracting. (Not to scale.)



- 21 An 11 cm square has a triangle cut from one corner, with shorter sides of 4 cm and 4 cm. Find the area left.

- 22 $1 \div 0.04$

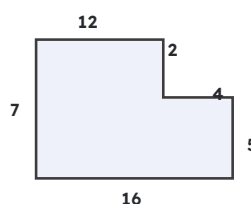
from 7.08

- 23 A 14 m × 10 m garden has a 12 m × 6 m pond. The rest is lawn. Find the lawn's area.

- 24 Find the area of a parallelogram with base 10 m and height 10 m.

from 8.07

- 25 Find the area two ways: by adding and by subtracting. (Not to scale.)



Challenging

Questions 26–32

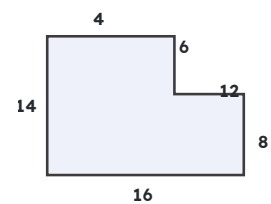
- 26 A 12 cm square has a triangle cut from one corner, with shorter sides of 6 cm and 6 cm. Find the area left.

- 27 A 10 m × 12 m garden has an 8 m × 8 m pond. The rest is lawn. Find the lawn's area.

- 28 A rhombus has a side of 13 cm. Find its perimeter.

from 6.10

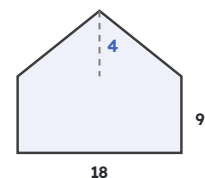
- 29 Find the area two ways: by adding and by subtracting. (Not to scale.)



- 30 Change 300 000 cm² to m².

from 8.04

- 31 Find the area of the shape. (Not to scale.)



- 32 An 8 cm square has a triangle cut from one corner, with shorter sides of 4 cm and 4 cm. Find the area left.

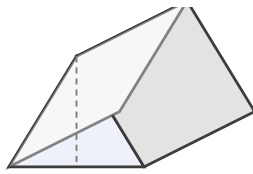
NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

A prism has the same cross-section all along its length and two identical ends. It is named after its cross-section. A solid can be drawn from the front, the side and the top.

E1 Name the prism.

E2 How many faces, edges and vertices does a rectangular prism have?

E3 Is a square pyramid a prism? Explain.

Easy

Questions 1–9

1 A prism has a cross-section that is a rectangle. Name the prism.

2 How many faces does a triangular prism have?

3 Is a sphere a prism?

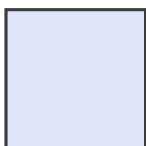
4 A prism has a cross-section that is a pentagon. Name the prism.

5 How many faces does a hexagonal prism have?

6 Is a cone a prism?

7 A prism has a cross-section that is a hexagon. Name the prism.

8 How many faces does a cube have?

9 Name the quadrilateral.


from 6.08

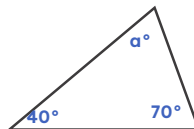
Easy

Questions 10–17

10 How many faces does a pentagonal prism have?

11 Is a cylinder a prism?

12 A prism has a cross-section that is a triangle. Name the prism.

13 Find α .


from 6.06

14 Find the number of faces, edges and vertices of a hexagonal prism.

faces: _____

vertices: ____ × ____ = ____

15 How many faces does a rectangular prism have?

16 Is a square pyramid a prism?

17 A 12 cm × 9 cm card has a 1 cm × 2 cm hole cut out. Find the area left.

from 8.08

Medium

Questions 18–24

18 Find the number of faces, edges and vertices of an octagonal prism.

19 Check Euler's rule $F + V - E = 2$ for a rectangular prism.

20 Add 8 m and 396 cm. Give the answer in metres.

from 8.01

21 A box is 3 cm long, 1 cm deep and 2 cm high. Describe its side view.

22 Find the number of faces, edges and vertices of a triangular prism.

23 A parallelogram has an area of 70 m² and a height of 5 m. Find its base.

from 8.07

24 Check Euler's rule $F + V - E = 2$ for a triangular prism.

Medium

Questions 25–32

- 25 Check Euler's rule $F + V - E = 2$ for a hexagonal prism.

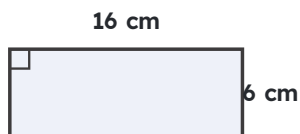
- 26 A box is 3 cm long, 1 cm deep and 2 cm high. Describe its top view.

- 27 A number plus 6 is 25. What is the number?

from 5.13

- 28 Find the number of faces, edges and vertices of a pentagonal prism.

- 29 Find the perimeter.

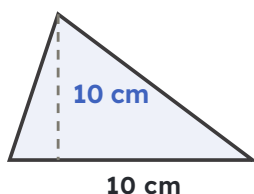


from 8.02

- 30 Draw the top view of a square pyramid.

- 31 Check Euler's rule $F + V - E = 2$ for a pentagonal prism.

- 32 Find the area. (Not to scale.)



from 8.06

Challenging

Questions 33–40

- 33 Draw the front (looking at a triangle end) view of a triangular prism lying on a rectangular face.

- 34 A prism has 13 faces. What shape is its cross-section, and how many edges does it have?

- 35 A triangle has angles 50° , 105° and 25° . Classify it by its angles and by its sides.

from 6.05

- 36 A prism has 12 vertices. Name it.

- 37 A shape is made of a $4\text{ m} \times 9\text{ m}$ rectangle and a $2\text{ m} \times 7\text{ m}$ rectangle. Find its area.

from 8.08

- 38 Draw the front view of a cylinder standing up.

- 39 A prism has 11 faces. What shape is its cross-section, and how many edges does it have?

- 40 A bike wheel has a diameter of 56 cm. How far does it roll in one turn (1 decimal place)?

from 8.03

Challenging

Questions 41–48

- 41 A prism has 10 faces. What shape is its cross-section, and how many edges does it have?

- 42 Draw the top view of a triangular prism lying on a rectangular face.

- 43 How many axes of symmetry does the capital letter C have?

from 6.03

- 44 A prism has 12 faces. What shape is its cross-section, and how many edges does it have?

- 45 Change $170\,000\text{ cm}^2$ to m^2 .

from 8.04

- 46 A prism has 9 faces. What shape is its cross-section, and how many edges does it have?

- 47 A prism has 14 faces. What shape is its cross-section, and how many edges does it have?

- 48 A rectangle is 10 cm by 9 cm. A diagonal cuts it into two triangles. Find the area of one triangle.

from 8.06

NAME _____ CLASS _____ DATE _____

WE DO
KEY IDEA
 $1 \text{ cm}^3 = 1000 \text{ mm}^3$ ($10 \times 10 \times 10$). $1 \text{ m}^3 = 1\,000\,000 \text{ cm}^3$ ($100 \times 100 \times 100$). A 1 cm^3 space holds 1 mL, and 1 m^3 holds 1000 L.

Watch your teacher.
Copy every step.

E1 Change 5 cm^3 to mm^3 .

E2 Change 2.5 m^3 to cm^3 .

E3 How many litres does 3 m^3 hold?

Easy

Questions 1–9

1 Change 11 cm^3 to mm^3 .

2 How many millilitres does 126 cm^3 hold?

3 How many litres does 4 m^3 hold?

4 Change 9 cm^3 to mm^3 .

5 How many millilitres does 743 cm^3 hold?

6 How many litres does 2 m^3 hold?

7 Change 16 cm^3 to mm^3 .

8 How many millilitres does 565 cm^3 hold?

9 Change $200\,000 \text{ cm}^2$ to m^2 .

from 8.04

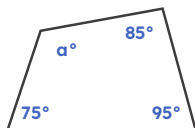
Easy

Questions 10–18

10 How many millilitres does 153 cm^3 hold?

11 How many litres does 9 m^3 hold?

12 Change 29 cm^3 to mm^3 .

13 Find α . (Not to scale.)


from 6.09

14 Change $18\,000 \text{ mm}^3$ to cm^3 .

 $\div$ ____

 $=$ ____ cm^3
15 How many millilitres does 755 cm^3 hold?

16 How many litres does 3 m^3 hold?

17 Change 25 t to kg .

from 8.01

18 Change 3 cm^3 to mm^3 .

Medium

Questions 19–26

19 Change $6\,000 \text{ mm}^3$ to cm^3 .

20 Change 4.5 m^3 to cm^3 .

21 Find the perimeter of a regular octagon with sides of 7 cm .

from 8.02

22 Change $2\,000 \text{ cm}^3$ to litres.

23 Change $43\,000 \text{ mm}^3$ to cm^3 .

24 Check Euler's rule $F + V - E = 2$ for a hexagonal prism.

from 8.09

25 Change 8.5 m^3 to cm^3 .

26 Change $2\,500 \text{ cm}^3$ to litres.

Medium

Questions 27–37

- 27 Change 9.5 m^3 to cm^3 .

- 28 Change 200 cm^3 to litres.

- 29 Round 53.9 to the nearest whole number.

from 7.10

- 30 Change $13\,000 \text{ mm}^3$ to cm^3 .

- 31 A circle has a radius of 3 m. What is its diameter?

from 8.03

- 32 Which is larger: 0.9 m^3 or $40\,000 \text{ cm}^3$?

- 33 Change 7 m^3 to cm^3 .

- 34 An $8 \text{ cm} \times 10 \text{ cm}$ card has a $3 \text{ cm} \times 3 \text{ cm}$ hole cut out. Find the area left.

from 8.08

- 35 Change $2\,300 \text{ cm}^3$ to litres.

- 36 A parallelogram has an area of 154 m^2 and a height of 14 m. Find its base.

from 8.07

- 37 Change $50\,000 \text{ mm}^3$ to cm^3 .

Challenging

Questions 38–47

- 38 Which is larger: 0.9 m^3 or $70\,000 \text{ cm}^3$?

- 39 A cube-shaped tank has sides of 20 cm. How many litres does it hold?

- 40 A triangle has angles 100° , 20° and 60° . Classify it by its angles and by its sides.

from 6.05

- 41 A pool holds 3.6 m^3 of water. How many litres is that?

- 42 How many mm^2 are in 1 cm^2 ?

from 8.04

- 43 Which is larger: 0.6 m^3 or $60\,000 \text{ cm}^3$?

- 44 A cube-shaped tank has sides of 40 cm. How many litres does it hold?

- 45 Find the area of a 7.5 m by 9 m rectangle.

from 8.05

- 46 A pool holds 5.3 m^3 of water. How many litres is that?

- 47 Change 3 h 6 min to minutes.

from 8.01

Challenging

Questions 48–56

- 48 A cube-shaped tank has sides of 30 cm. How many litres does it hold?

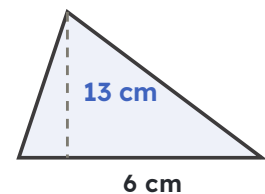
- 49 A pool holds 5.2 m^3 of water. How many litres is that?

- 50 0.2×0.9

from 7.06

- 51 Which is larger: 0.6 m^3 or $30\,000 \text{ cm}^3$?

- 52 Find the area. (Not to scale.)



from 8.06

- 53 Change 700 cm^3 to litres.

- 54 A cube-shaped tank has sides of 60 cm. How many litres does it hold?

- 55 Is a cone a prism?

from 8.09

- 56 A pool holds 2.7 m^3 of water. How many litres is that?

8.11 Volume of a rectangular prism

YEAR 7 · CHAPTER 8: AREA AND VOLUME · PAGE 1 OF 2

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

Volume of a rectangular prism = length $\times$ width $\times$ height. Volume is measured in cubic units, such as cm^3 and m^3 .

E1 Find the volume of a box 5 cm long, 4 cm wide and 3 cm high.

E2 Find the volume of a cube with sides of 4 cm.

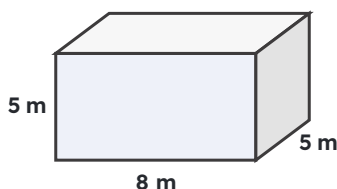
E3 A box has a volume of 120 cm^3 . It is 6 cm long and 5 cm wide. Find its height.

Easy

Questions 1–7

1 Find the volume of a box 9 cm long, 9 cm wide and 4 cm high.

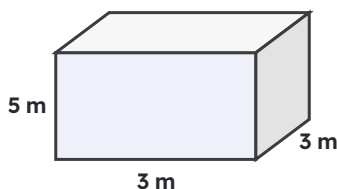
2 Find the volume.



3 Find the volume of a cube with sides of 7 cm.

4 Find the volume of a box 12 cm long, 2 cm wide and 3 cm high.

5 Find the volume.



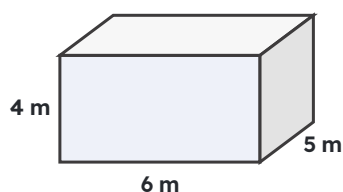
6 Find the volume of a cube with sides of 8 cm.

7 Find the volume of a box 11 cm long, 2 cm wide and 7 cm high.

Easy

Questions 8–12

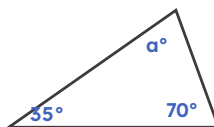
8 Find the volume.



9 Find the volume of a cube with sides of 10 cm.

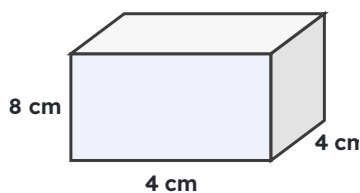
10 Find the volume of a box 10 cm long, 9 cm wide and 2 cm high.

11 Find α .



from 6.06

12 Find the volume. Write the formula first.



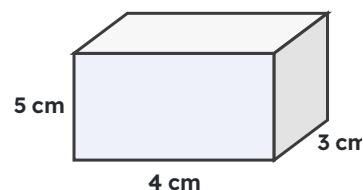
$$V = ___ \times ___ \times ___$$

$$= ___ \text{ cm}^3$$

Medium

Questions 13–17

13 Find the volume. Write the formula first.



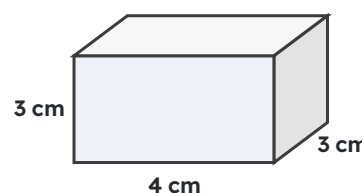
14 A box has a volume of 110 cm^3 . It is 11 cm long and 5 cm wide. Find its height.

15 Add 9 m and 915 cm. Give the answer in metres.

from 8.01

16 Find the volume of a box 2.3 m by 2 m by 4 m.

17 Find the volume. Write the formula first.



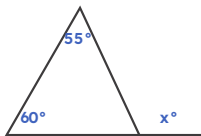
Medium

Questions 18–24

- 18 A box has a volume of 60 cm^3 . It is 6 cm long and 2 cm wide. Find its height.

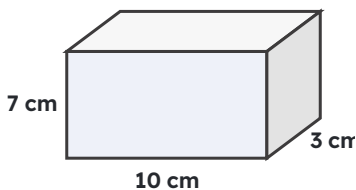
- 19 Find the volume of a box 2.3 m by 4 m by 3 m.

- 20 Find x , the exterior angle.

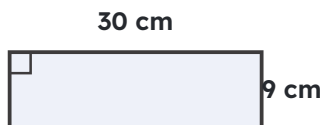


from 6.07

- 21 Find the volume. Write the formula first.



- 22 Find the perimeter.



from 8.02

- 23 A fish tank is 80 cm by 30 cm by 20 cm. How many litres does it hold?

- 24 A box has a volume of 27 cm^3 . It is 3 cm long and 3 cm wide. Find its height.

Challenging

Questions 25–32

- 25 A fish tank is 30 cm by 20 cm by 40 cm. How many litres does it hold?

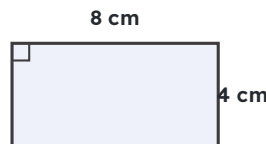
- 26 How many 2 cm cubes fit in a box 6 cm by 8 cm by 8 cm?

- 27 Solve $\frac{y}{2} + 6 = 13$.

from 5.12

- 28 A cube's sides are doubled from 4 cm to 8 cm. How many times larger is the volume?

- 29 Find the area.



from 8.05

- 30 A fish tank is 70 cm by 60 cm by 20 cm. How many litres does it hold?

- 31 How many 2 cm cubes fit in a box 10 cm by 8 cm by 4 cm?

- 32 A bike wheel has a diameter of 59 cm. How far does it roll in one turn (1 decimal place)?

from 8.03

Challenging

Questions 33–42

- 33 How many 2 cm cubes fit in a box 8 cm by 8 cm by 8 cm?

- 34 A cube's sides are doubled from 3 cm to 6 cm. How many times larger is the volume?

- 35 A rhombus has a side of 8 cm. Find its perimeter.

from 6.10

- 36 A fish tank is 30 cm by 40 cm by 30 cm. How many litres does it hold?

- 37 Change $250\,000 \text{ cm}^2$ to m^2 .

from 8.04

- 38 Find the volume of a box 2.3 m by 4 m by 4 m.

- 39 How many 2 cm cubes fit in a box 8 cm by 10 cm by 6 cm?

- 40 A shape is made of a 4 m $\times$ 5 m rectangle and a 10 m $\times$ 5 m rectangle. Find its area.

from 8.08

- 41 A cube's sides are doubled from 2 cm to 4 cm. How many times larger is the volume?

- 42 Change $90\,000 \text{ mm}^3$ to cm^3 .

from 8.10

NAME _____ CLASS _____ DATE _____

WE DO

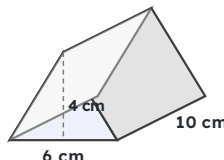
Watch your teacher.
Copy every step.

KEY IDEA

The volume of any prism = the area of its cross-section $\times$ its length: $V = Ah$. Find the area of the end first, then multiply by the length.

- E1** A prism has a cross-section of 15 cm^2 and a length of 8 cm. Find its volume.

- E2** A triangular prism has a triangle of base 6 cm and height 4 cm, and is 10 cm long. Find its volume.



- E3** A prism has a volume of 96 m^3 and a cross-section of 12 m^2 . How long is it?

Easy

Questions 1–8

- 1** A prism has a cross-section of 32 cm^2 and a length of 6 cm. Find its volume.

- 2** Find the area of the triangular end: base 6 cm, height 4 cm.

- 3** A prism has a cross-section of 24 cm^2 and a length of 9 cm. Find its volume.

- 4** Find the area of the triangular end: base 10 cm, height 8 cm.

- 5** A prism has a cross-section of 21 cm^2 and a length of 10 cm. Find its volume.

- 6** Find the area of the triangular end: base 8 cm, height 4 cm.

- 7** A prism has a cross-section of 22 cm^2 and a length of 13 cm. Find its volume.

- 8** Find the area of the triangular end: base 4 cm, height 5 cm.

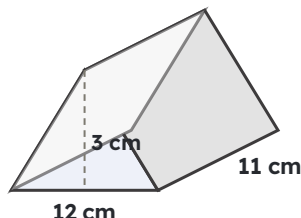
Easy

Questions 9–13

- 9** Find the area of the triangular end: base 6 cm, height 3 cm.

- 10** A prism has a cross-section of 40 cm^2 and a length of 8 cm. Find its volume.

- 11** Find the volume of the triangular prism.



$$A = \frac{1}{2} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

$$V = \underline{\hspace{2cm}}$$

- 12** Find x , the exterior angle.



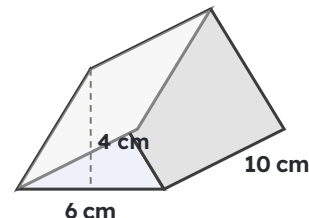
from 6.07

- 13** Find the area of the triangular end: base 4 cm, height 4 cm.

Medium

Questions 14–17

- 14** Find the volume of the triangular prism.

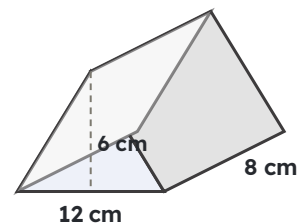


- 15** A prism has a volume of 378 m^3 and a cross-section of 27 m^2 . How long is it?

- 16** Add 5 m and 490 cm. Give the answer in metres.

from 8.01

- 17** Find the volume of the triangular prism.

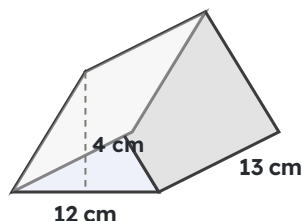


Medium

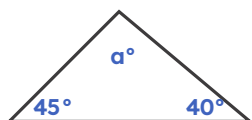
Questions 18–23

- 18 A prism has a volume of 91 m^3 and a cross-section of 7 m^2 . How long is it?

- 19 Find the volume of the triangular prism.

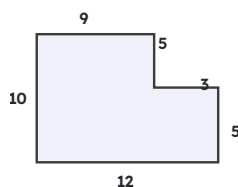


- 20 Find α .

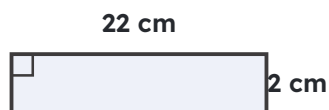


from 6.06

- 21 A prism has the L-shaped cross-section shown and is 6 cm long. Find its volume. (Not to scale.)



- 22 Find the perimeter.



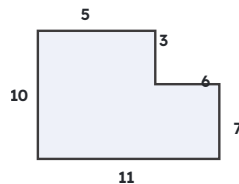
from 8.02

- 23 A prism has a volume of 55 m^3 and a cross-section of 5 m^2 . How long is it?

Challenging

Questions 24–29

- 24 A prism has the L-shaped cross-section shown and is 3 cm long. Find its volume. (Not to scale.)



- 25 A tent is a triangular prism. Its end is a triangle with base 3 m and height 2 m, and it is 4 m long. Find its volume.

- 26 Given $80 \div 4 = 20$, find $8 \div 4$.

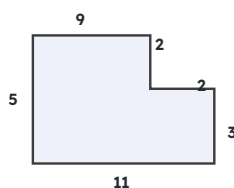
from 7.05

- 27 Two prisms have the same cross-section, 15 cm^2 . One is 3 cm long and the other is 9 cm long. Compare their volumes.

- 28 A rectangle is 6 cm by 9 cm. A diagonal cuts it into two triangles. Find the area of one triangle.

from 8.06

- 29 A prism has the L-shaped cross-section shown and is 5 cm long. Find its volume. (Not to scale.)



Challenging

Questions 30–37

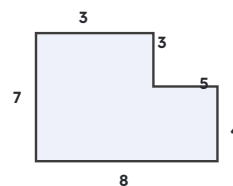
- 30 A tent is a triangular prism. Its end is a triangle with base 4 m and height 2 m, and it is 3 m long. Find its volume.

- 31 Two prisms have the same cross-section, 18 cm^2 . One is 4 cm long and the other is 12 cm long. Compare their volumes.

- 32 Write $\frac{1}{9}$ as a decimal. Use a dot for the repeating digit.

from 7.09

- 33 A prism has the L-shaped cross-section shown and is 10 cm long. Find its volume. (Not to scale.)



- 34 Change $130\,000 \text{ m}^2$ to ha.

from 8.04

- 35 A prism has a volume of 112 m^3 and a cross-section of 16 m^2 . How long is it?

- 36 A tent is a triangular prism. Its end is a triangle with base 4 m and height 3 m, and it is 5 m long. Find its volume.

- 37 Is a square pyramid a prism?

from 8.09

NAME _____ CLASS _____ DATE _____

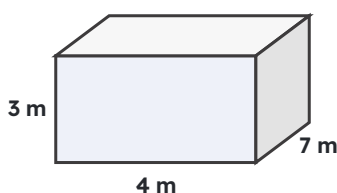
WE DO**KEY IDEA**Capacity is how much a container holds. 1 L = 1000 mL. A space of 1 cm³ holds 1 mL, so 1000 cm³ holds 1 L, and 1 m³ holds 1000 L.Watch your teacher.
Copy every step.**E1** Change 2.75 L to millilitres.

E2 A box is 20 cm by 10 cm by 15 cm. How many litres does it hold?

E3 A 2 000 L tank fills at 25 L a minute. How long does it take to fill?

Easy

Questions 1–9

1 Change 4.3 L to millilitres.**2** Change 4 398 mL to litres.**3** A container holds 7 cm³. What is its capacity in millilitres?**4** Change 2.7 L to millilitres.**5** Change 3 539 mL to litres.**6** A container holds 449 cm³. What is its capacity in millilitres?**7** Change 8.1 L to millilitres.**8** Change 1 538 mL to litres.**9** Find the volume.

from 8.11

Easy

Questions 10–18

10 Change 581 mL to litres.**11** A container holds 657 cm³. What is its capacity in millilitres?**12** Change 4.4 L to millilitres.**13** Find x , the exterior angle.

from 6.07

14 A box is 50 cm by 10 cm by 15 cm. How many litres does it hold? $V =$ _____ $=$ _____ L**15** Change 218 mL to litres.**16** A container holds 683 cm³. What is its capacity in millilitres?**17** How many millilitres does 133 cm³ hold?

from 8.10

18 Change 7.4 L to millilitres.**Medium**

Questions 19–24

19 A box is 50 cm by 20 cm by 5 cm. How many litres does it hold?**20** How many 500 mL bottles can be filled from a 6 L drum?**21** Add 9 m and 574 cm. Give the answer in metres.

from 8.01

22 A box is 40 cm by 10 cm by 5 cm. How many litres does it hold?**23** How many 250 mL bottles can be filled from an 8 L drum?**24** A prism has a volume of 68 m³ and a cross-section of 17 m². How long is it?

from 8.12

Medium

Questions 25–33

- 25 How many 250 mL bottles can be filled from a 5 L drum?

- 26 A box is 30 cm by 10 cm by 15 cm. How many litres does it hold?

- 27 Write $\frac{1}{3}$ as a decimal. Use a dot for the repeating digit.

from 7.09

- 28 A 1 200 L tank fills at 40 L a minute. How long does it take to fill?

- 29 Find the perimeter.



from 8.02

- 30 How many 500 mL bottles can be filled from a 4 L drum?

- 31 A box is 20 cm by 10 cm by 15 cm. How many litres does it hold?

- 32 Find the volume of a cube with sides of 5 cm.

from 8.11

- 33 A 1 900 L tank fills at 10 L a minute. How long does it take to fill?

Challenging

Questions 34–41

- 34 A 700 L tank fills at 20 L a minute. How long does it take to fill?

- 35 A pool is 2 m by 3 m by 1 m deep. How many litres does it hold?

- 36 \$1.75, \$8.65 and \$6.75 are paid with a \$50 note. Find the change.

from 7.11

- 37 Water is poured into a 50 cm by 20 cm tank until it is 3 cm deep. How many litres is that?

- 38 A prism has a cross-section of 34 cm^2 and a length of 14 cm. Find its volume.

from 8.12

- 39 A 1 800 L tank fills at 15 L a minute. How long does it take to fill?

- 40 A pool is 3 m by 1 m by 1 m deep. How many litres does it hold?

- 41 A bike wheel has a diameter of 75 cm. How far does it roll in one turn (1 decimal place)?

from 8.03

Challenging

Questions 42–51

- 42 A pool is 5 m by 3 m by 2 m deep. How many litres does it hold?

- 43 Water is poured into a 60 cm by 40 cm tank until it is 4 cm deep. How many litres is that?

- 44 $0.8 \div 0.4$

from 7.08

- 45 A 1 500 L tank fills at 10 L a minute. How long does it take to fill?

- 46 Change $80\,000 \text{ cm}^2$ to m^2 .

from 8.04

- 47 How many 500 mL bottles can be filled from a 9 L drum?

- 48 A pool is 4 m by 1 m by 2 m deep. How many litres does it hold?

- 49 Change 14 cm^3 to mm^3 .

from 8.10

- 50 Water is poured into a 50 cm by 30 cm tank until it is 2 cm deep. How many litres is that?

- 51 A prism has a volume of 104 m^3 and a cross-section of 26 m^2 . How long is it?

from 8.12